

Program 8 – Check Whether a Number is Prime

Answer / Python Program

```
def prime(n):
    if n <= 1:
        return False

    for i in range(2, n):
        if n % i == 0:
            return False

    return True

num = int(input("Enter a number: "))
if prime(num):
    print(num, "is a Prime Number")
else:
    print(num, "is Not a Prime Number")
```

Output

```
Enter a number: 17
17 is a Prime Number
```

Note: The programs were formatted as working Python programs. The triangle program uses the complete area formula $\frac{1}{2} \times \text{base} \times \text{height}$, and the maximum-number program avoids recursive shadowing of Python's built-in `max()`.